Problem 1

Let

$$|\psi\rangle = a\,|0\rangle_A + b\,|1\rangle_A\,, \qquad |\phi\rangle = c\,|0\rangle_B + d\,|1\rangle_B\,,$$
$$|a|^2 + |b|^2 = 1 \quad |c|^2 + |d|^2 = 1$$

Note that

$$|\psi,\phi\rangle = |\psi\rangle \otimes |\phi\rangle = ac\,|00\rangle + ad\,|01\rangle + bc\,|10\rangle + bd\,|11\rangle$$

Starting from this

$$\left\langle \Phi_+^{(AB)} | \psi,\phi \right\rangle = \frac{1}{\sqrt{2}}\left(\langle 00| + \langle 11|\right)\left(ac\,|00\rangle + ad\,|01\rangle + bc\,|10\rangle + bd\,|11\rangle\right)$$
$$= \frac{1}{\sqrt{2}}\Big[ac\,\langle 00|00\rangle + \cancel{ad\,\langle 00|01\rangle} + \cancel{bc\,\langle 00|10\rangle} + \cancel{bd\,\langle 00|11\rangle}$$
$$+ \cancel{ac\,\langle 11|00\rangle} + \cancel{ad\,\langle 11|01\rangle} + \cancel{bc\,\langle 11|10\rangle} + bd\,\langle 11|11\rangle\Big]$$
$$= \frac{1}{\sqrt{2}}\,(ac + bd)$$

Applying Cauchy-Schwarz

$$|ac + bd|^2 \le \left(|a|^2 + |b|^2\right)\left(|c|^2 + |d|^2\right) = 1 \cdot 1 = 1$$

Therefore $|ac + bd| \le 1$, and hence

$$\left|\left\langle \Phi_+^{(AB)} | \psi,\phi \right\rangle\right| = \frac{|ac + bd|}{\sqrt{2}} \le \frac{1}{\sqrt{2}} < 1$$

So $\left|\left\langle \Phi_+^{(AB)} | \psi,\phi \right\rangle\right|$ never can be equal to 1 so it is not a product state, it is an entangled state.

Problem 2

Problem 2a

$A = A$ always, so the probability not being itself is zero.

$$d(A, A) = \Pr(A \ne A) = 0$$

Probability of two things are commutative so $Pr(a, b) = Pr(b, a)$ so

$$d(A, B) = d(B, A)$$

We got to think of it using set theory so if $A \ne C$ then either $A \ne B$ or $B \ne C$, so $\{A \ne C\} \subseteq \{A \ne B\} \cup \{B \ne C\}$. By the union bound

$$d(A, C) \le d(A, B) + d(B, C)$$

Problem 2b

$A, B \in \{-1, +1\}$ gives $AB = +1$ if $A = B$, $AB = -1$ if $A \ne B$, so

$$\mathbb{E}[AB] = \Pr(A = B) - \Pr(A \ne B) = 1 - 2\,d(A, B)$$
$$\Rightarrow \quad d(A, B) = \frac{1 - \mathbb{E}[AB]}{2}$$

Problem 2c

From 2a

$$d(A_1, A_2) \leq d(A_1, B_1) + d(B_1, A_2)$$
$$d(A_1, B_2) \leq d(A_1, A_2) + d(A_2, B_2)$$

Substitute

$$d(A_1, B_2) \leq d(A_1, B_1) + d(B_1, A_2) + d(A_2, B_2)$$

This must be true. So we know

$$A_1 \ \to^{d(A_1,B_1)} \ B_1 \ \to^{d(B_1,A_2)} \ A_2 \ \to^{d(A_2,B_2)} \ B_2$$

That means A_1B_2 is the diagonal. Substituting $d = (1 - \mathbb{E}[XY])/2$ from Problem 2b recovers

$$\mathbb{E}[A_1B_2] - \mathbb{E}[A_1B_1] - \mathbb{E}[B_1A_2] - \mathbb{E}[A_2B_2] \leq 2$$

Problem 2d

Let $|\Psi_-\rangle$,

$$\mathbb{E}[(\vec{\sigma} \cdot \hat{a}) \otimes (\vec{\sigma} \cdot \hat{b})] = -\cos\theta_{ab} \quad d(A, B) = \frac{1 + \cos\theta_{ab}}{2}$$

Take $\hat{a}_1, \hat{b}_1, \hat{a}_2, \hat{b}_2$ on the same plane at angles

$$\alpha_{a_1} = 0 \quad \alpha_{b_1} = \tfrac{3\pi}{4} \quad \alpha_{a_2} = \tfrac{3\pi}{2} \quad \alpha_{b_2} = \tfrac{\pi}{4}$$

The angle between two coplanar unit vectors is $\theta = \min(|\Delta\alpha|, 2\pi - |\Delta\alpha|)$, so

$$\theta_{a_1b_1} = |0 - \tfrac{3\pi}{4}| = \tfrac{3\pi}{4}$$

$$\theta_{b_1a_2} = |\tfrac{3\pi}{4} - \tfrac{3\pi}{2}| = |\tfrac{3\pi}{4} - \tfrac{6\pi}{4}| = \tfrac{3\pi}{4}$$
$$\theta_{a_2b_2} = 2\pi - |\tfrac{3\pi}{2} - \tfrac{\pi}{4}| = 2\pi - |\tfrac{6\pi}{4} - \tfrac{\pi}{4}| = 2\pi - \tfrac{5\pi}{4} = \tfrac{3\pi}{4}$$
$$\theta_{a_1b_2} = |0 - \tfrac{\pi}{4}| = \tfrac{\pi}{4}$$

Plug into $d(A, B) = \frac{1+\cos\theta_{ab}}{2}$, using $\cos\frac{3\pi}{4} = -\frac{1}{\sqrt{2}}$ and $\cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$

$$d(A_1, B_1) = \frac{1 + \cos(3\pi/4)}{2} = \frac{1 + (-1/\sqrt{2})}{2} = \frac{1 - 1/\sqrt{2}}{2}$$

$$d(B_1, A_2) = \frac{1 + \cos(3\pi/4)}{2} = \frac{1 - 1/\sqrt{2}}{2}$$

$$d(A_2, B_2) = \frac{1 + \cos(3\pi/4)}{2} = \frac{1 - 1/\sqrt{2}}{2}$$

$$d(A_1, B_2) = \frac{1 + \cos(\pi/4)}{2} = \frac{1 + 1/\sqrt{2}}{2}$$

From 2c,

$$d(A_1, B_2) \leq d(A_1, B_1) + d(B_1, A_2) + d(A_2, B_2)$$
$$0.85 \approx \frac{1 + 1/\sqrt{2}}{2} \not\leq \frac{3 - 3/\sqrt{2}}{2} \approx 0.44$$

Hence contradiction, so it is not possible.

Problem 3

Problem 3a

σ_Z has eigenvectors $+1$ for $|0\rangle$ and -1 for $|1\rangle$ so

$$|\text{GHZ}\rangle = \frac{1}{\sqrt{2}}(|0\rangle_A |00\rangle_{BC} - |1\rangle_A |11\rangle_{BC})$$

Alice measures $+1 \Rightarrow$ project onto $|0\rangle_A$

$$|\text{GHZ}\rangle \to |0\rangle_A |00\rangle_{BC} = |0\rangle_A \otimes |0\rangle_B \otimes |0\rangle_C$$

Alice measures $-1 \Rightarrow$ project onto $|1\rangle_A$

$$|\text{GHZ}\rangle \to -|1\rangle_A |11\rangle_{BC} = -|1\rangle_A \otimes |1\rangle_B \otimes |1\rangle_C$$

Either way the BC subsystem is a product of single-qubit states.

Problem 3b

σ_X has eigenvectors $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$, so

$$|0\rangle = \frac{|+\rangle + |-\rangle}{\sqrt{2}} \qquad |1\rangle = \frac{|+\rangle - |-\rangle}{\sqrt{2}}$$

Sub into $|\text{GHZ}\rangle$

$$\begin{aligned}|\text{GHZ}\rangle &= \frac{1}{\sqrt{2}}\left[\frac{|+\rangle_A + |-\rangle_A}{\sqrt{2}}|00\rangle_{BC} - \frac{|+\rangle_A - |-\rangle_A}{\sqrt{2}}|11\rangle_{BC}\right] \\ &= \frac{1}{2}\Big[\,|+\rangle_A (|00\rangle - |11\rangle)_{BC} + |-\rangle_A (|00\rangle + |11\rangle)_{BC}\Big]\end{aligned}$$

Alice measures $+1 \Rightarrow \text{BC} = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) = \left|\Phi_-^{(BC)}\right\rangle$.

Alice measures $-1 \Rightarrow \text{BC} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \left|\Phi_+^{(BC)}\right\rangle$.

Both are Bell states. By Schwartz like in 1),

$$\left|\left\langle \Phi_\pm^{(BC)} | \psi, \phi \right\rangle\right| = \frac{|ac \pm bd|}{\sqrt{2}} \le \frac{1}{\sqrt{2}} < 1$$

But if BC were $|\psi\rangle \otimes |\phi\rangle$ the overlap would equal 1.

Contradiction thus BC is entangled.

Problem 4

Problem 4a

For arbitrary $|\psi\rangle, |\phi\rangle, |\psi'\rangle, |\phi'\rangle$

$$\begin{aligned}\langle \psi', \phi'| (A \otimes B)^+ |\psi, \phi\rangle &= \langle \psi, \phi | A \otimes B | \psi', \phi'\rangle^* \\ = \big(\langle \psi| A |\psi'\rangle \langle \phi| B |\phi'\rangle\big)^* &= \langle \psi| A |\psi'\rangle^* \langle \phi| B |\phi'\rangle^* \\ = \langle \psi'| A^+ |\psi\rangle \langle \phi'| B^+ |\phi\rangle &= \langle \psi', \phi'| A^+ \otimes B^+ |\psi, \phi\rangle\end{aligned}$$

Matrix elements agree on all states, so

$$(A \otimes B)^+ = A^+ \otimes B^+$$

Problem 4b

$A^+ = A$, $B^+ = B$, so by 4a

$$(A \otimes B)^+ = A^+ \otimes B^+ = A \otimes B$$

Problem 4c

Use $(X \otimes Y)(X' \otimes Y') = (XX') \otimes (YY')$ and 4a throughout

Positive: $A \triangleq M^+M$, $B \triangleq N^+N$, so

$$A \otimes B = (M^+M) \otimes (N^+N) = (M \otimes N)^+(M \otimes N) \geq 0$$

Unitary: $A^+A = I$, $B^+B = I$, so

$$(A \otimes B)^+(A \otimes B) = (A^+ \otimes B^+)(A \otimes B) = (A^+A) \otimes (B^+B) = I \otimes I = I$$

Normal: $A^+A = AA^+$, $B^+B = BB^+$, so

$$(A \otimes B)^+(A \otimes B) = (A^+A) \otimes (B^+B) = (AA^+) \otimes (BB^+) = (A \otimes B)(A \otimes B)^+$$

Problem 4d

No. Take $A = B = iI$. Neither is hermitian ($A^+ = -iI \neq A$), but by 4a

$$(A \otimes B)^+ = A^+ \otimes B^+ = (-iI) \otimes (-iI) = -I \otimes I = A \otimes B$$

which is hermitian. More generally, anti-herm$\otimes$ anti-herm is herm

$$(A \otimes B)^+ = (-A) \otimes (-B) = A \otimes B$$

Problem 5

Problem 5a

R is unitary, so preserves inner products

$$\langle R0|R1\rangle = \langle 0|1\rangle = 0$$

Hence $R\,|1\rangle \perp R\,|0\rangle = |n_+\rangle$. In a 2D Hilbert space there is exactly one direction orthogonal to $|n_+\rangle$, namely $|n_-\rangle$ (eigenvectors are unique only up to a phase), so

$$R\,|1\rangle = e^{i\beta}\,|n_-\rangle$$

Problem 5b

Allow arbitrary phases $R\,|0\rangle = e^{i\alpha}\,|n_+\rangle$, $R\,|1\rangle = e^{i\beta}\,|n_-\rangle$. Apply $R \otimes R$ to $|\Psi_-\rangle$

$$\begin{aligned}(R \otimes R)\,|\Psi_-\rangle &= \frac{1}{\sqrt{2}}\big[(R\,|0\rangle) \otimes (R\,|1\rangle) - (R\,|1\rangle) \otimes (R\,|0\rangle)\big] \\ &= \frac{e^{i(\alpha+\beta)}}{\sqrt{2}}\big[\,|n_+\rangle \otimes |n_-\rangle - |n_-\rangle \otimes |n_+\rangle\,\big]\end{aligned}$$

Expand $|n_+\rangle = c\,|0\rangle + s\,|1\rangle$ and $|n_-\rangle = -s^*\,|0\rangle + c\,|1\rangle$ with $c = \cos(\theta/2)$, $s = e^{i\phi}\sin(\theta/2)$, $c^2 + |s|^2 = 1$

$$|n_+\rangle \otimes |n_-\rangle = -cs^*\,|00\rangle + c^2\,|01\rangle - |s|^2\,|10\rangle + cs\,|11\rangle$$
$$|n_-\rangle \otimes |n_+\rangle = -s^*c\,|00\rangle - |s|^2\,|01\rangle + c^2\,|10\rangle + cs\,|11\rangle$$

Subtract

$$|n_+\rangle \otimes |n_-\rangle - |n_-\rangle \otimes |n_+\rangle = (c^2 + |s|^2)\,|01\rangle - (|s|^2 + c^2)\,|10\rangle = |01\rangle - |10\rangle = \sqrt{2}\,|\Psi_-\rangle$$
$$\Rightarrow \quad (R \otimes R)\,|\Psi_-\rangle = e^{i(\alpha+\beta)}\,|\Psi_-\rangle$$

i.e. $|\Psi_-\rangle$ up to a phase.

Problem 5c

Let R be the rotation taking $\hat{z} \to \hat{n}$. Then $R^+ |n_\pm\rangle = |0/1\rangle$ (up to phase) and the eigenvalues match, so $R^+ S_n R = S_z$, i.e.

$$S_n = R\, S_z\, R^+$$

Therefore

$$\begin{aligned}\langle\Psi_-|\, S_n \otimes S_n\, |\Psi_-\rangle &= \langle\Psi_-|\,(R\, S_z\, R^+) \otimes (R\, S_z\, R^+)\,|\Psi_-\rangle \\ &= \langle\Psi_-|\,(R \otimes R)(S_z \otimes S_z)(R^+ \otimes R^+)\,|\Psi_-\rangle\end{aligned}$$

From (b), $(R^+ \otimes R^+)\,|\Psi_-\rangle = e^{-i(\alpha+\beta)}\,|\Psi_-\rangle$ and $\langle\Psi_-|\,(R \otimes R) = e^{i(\alpha+\beta)}\,\langle\Psi_-|$. The phases cancel out and we get

$$\langle\Psi_-|\, S_n \otimes S_n\, |\Psi_-\rangle = \langle\Psi_-|\, S_z \otimes S_z\, |\Psi_-\rangle$$

Compute the RHS using $S_z\,|0\rangle = \frac{\hbar}{2}\,|0\rangle$ and $S_z\,|1\rangle = -\frac{\hbar}{2}\,|1\rangle$

$$(S_z \otimes S_z)\,|01\rangle = -\tfrac{\hbar^2}{4}\,|01\rangle \qquad (S_z \otimes S_z)\,|10\rangle = -\tfrac{\hbar^2}{4}\,|10\rangle$$

$$(S_z \otimes S_z)\,|\Psi_-\rangle = \frac{1}{\sqrt{2}}\left[-\tfrac{\hbar^2}{4}\,|01\rangle + \tfrac{\hbar^2}{4}\,|10\rangle\right] = -\tfrac{\hbar^2}{4}\,|\Psi_-\rangle$$

$$\Rightarrow \quad \langle\Psi_-|\, S_z \otimes S_z\, |\Psi_-\rangle = -\tfrac{\hbar^2}{4}$$

$$\frac{4}{\hbar^2}\,\langle\Psi_-|\, S_n \otimes S_n\, |\Psi_-\rangle = \frac{4}{\hbar^2}\cdot\left(-\tfrac{\hbar^2}{4}\right) = -1$$

Problem 5d

DNE. If parallel measurements always got the same results, then $S_n \otimes S_n$ always returns $+\hbar^2/4$ (either both $+\hbar/2$ or both $-\hbar/2$), so

$$\langle\psi|\, S_n \otimes S_n\, |\psi\rangle = +\tfrac{\hbar^2}{4} \quad \forall \hat{n}$$

Sum over orthogonal axes $\hat{x}, \hat{y}, \hat{z}$

$$\langle\psi|\, S_x \otimes S_x\, |\psi\rangle + \langle\psi|\, S_y \otimes S_y\, |\psi\rangle + \langle\psi|\, S_z \otimes S_z\, |\psi\rangle = \tfrac{3\hbar^2}{4}$$

The LHS equals $\langle\psi|\, \vec{S}_1\vec{S}_2\, |\psi\rangle$. Use the total spin $\vec{S} = \vec{S}_1 + \vec{S}_2$

$$\vec{S}^2 = \vec{S}_1^2 + \vec{S}_2^2 + 2\,\vec{S}_1\vec{S}_2 = \tfrac{3\hbar^2}{2} + 2\,\vec{S}_1\vec{S}_2$$

For two spin-$\frac{1}{2}$ particles, $\vec{S}^2$ has eigenvalues 0 (singlet) or $2\hbar^2$ (triplet), so

$$\vec{S}_1\vec{S}_2 \in \left\{-\tfrac{3\hbar^2}{4},\ +\tfrac{\hbar^2}{4}\right\}$$

Max. eigenvalue is $\hbar^2/4$, so $\langle\psi|\, \vec{S}_1\vec{S}_2\, |\psi\rangle \le \hbar^2/4$ for any pure state (and $\mathrm{Tr}(\rho\, \vec{S}_1\vec{S}_2) \le \hbar^2/4$ for any mixed state). But the anti-singlet would need $\frac{3\hbar^2}{4} > \frac{\hbar^2}{4}$.

Contradiction.